Operating System Security

Week 4: Windows Access Control

Windows Protection System

Discretionary – for managing protection state, object labeling and protection domain transitions

More variety in objects and operations than UNIX, and additional flexibility in assigning them to subjects

Each process is assigned a token that describes the process's identity, consisting of:

- a UID, set of group SIDs, a set of alias SIDs to enable actions on behalf of another identity, and a set of ad-hoc privileges just associated with this token

Windows Access Control Lists

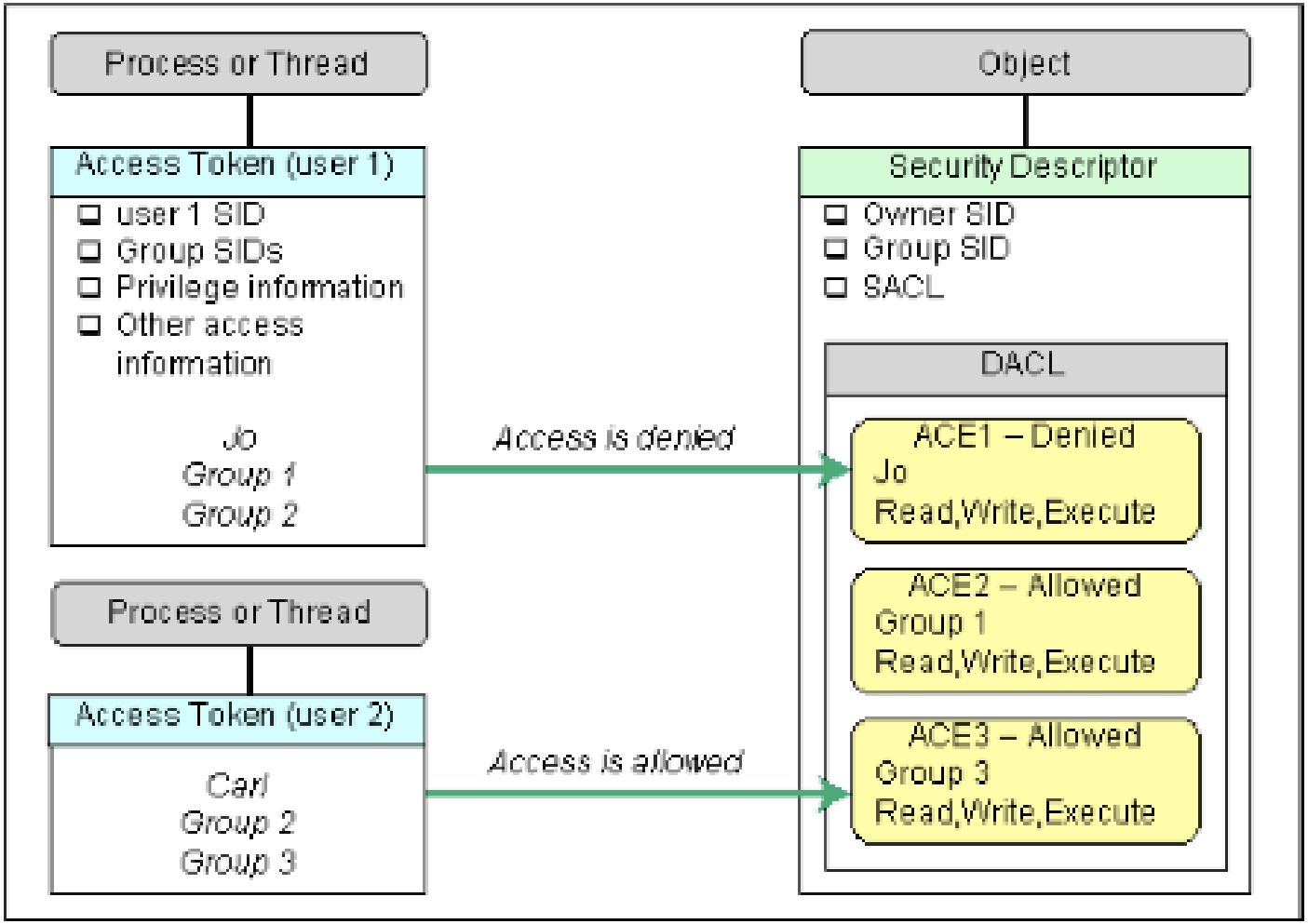
Windows supports arbitrary ACLs rather than the limited mode bits approach of UNIX

A Windows ACL stores a set of access control entries (ACEs) that describe which operations a SID (user, group, or alias) can perform on that object.

An ACE can either grant or deny an operation

The operations in an ACE are interpreted based on the object type of the target object

Windows Access Control



Share and NTFS Permissions

Each permission can be configured as Allow or Deny.

Deny overrides Allow if set at the same level.

Permissions can be set per file (except shares), per folder, per-user, and per-group.

Share permissions apply only when a resource is contacted over a network connection:

- Full Control
- Change
- Read

14 Special Permissions

Permission	Full Control	Modify	Read & Execute	List	Read	Write
Traverse Folder/ Execute File	X	X	X	X		
List Folder/Read Data	X	X	X	X	X	
Read Attributes	X	X	X	X	X	
Read Extended Attributes	X	X	X	X	X	
Create Files/Write Data	X	X				X
Create Folders/Append Data	X	X				X
Write Attributes	X	X				X
Write Extended Attributes	X	X				X
Delete Subfolders and Files	X					
Delete	X	X				
Read Permissions	X	X	X	X	X	X
Change Permissions	X					
Take Ownership	X					
Synchronize	X	X	X	X	X	X

Special Permissions

For information about these permissions and what they mean, see the following reference:

- Microsoft, “Permission Entry Dialog Box”, https://docs.microsoft.com/en-us/previous-versions/windows/it-pro/windows-server-2008-R2-and-2008/cc753992%28v%3dws.10%29 , last accessed Jan. 23, 2020.

What are the differences between the following?

- Read Permissions vs. Read Attributes vs. Read Data
- Read vs. Read & Execute
- Write vs. Read

Inheritance

- Inheritance – getting default permissions from parent folders.

- You can turn it on or off by going to the security tab
- ->Advanced -> change permissions
  - Then toggle the checkbox.

- You can check their effect by looking at “Effective Permissions” utility.

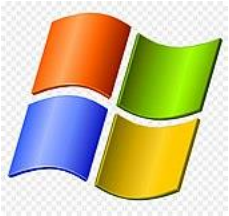
Windows and Permissions Inheritance – Some Odd Behaviour

NTFS is the standard file system of Windows NT, including its later versions Windows 2000, Windows XP, Windows Server 2003/2008/2012, Windows Vista/7/8/10.

Try it -> How IT works NTFS Permissions

- http://technet.microsoft.com/en-us/magazine/2005.11.howitworksntfs.aspx (last accessed Jan. 26, 2021).

Summary



- Windows uses ACLs
- ACLs can be inherited from parent folders, leading to unexpected behaviour if you are not careful!
- Much richer expression of permissions is possible than in a typical Unix/Linux environment.
  - But the downside of this is that they can be more difficult to understand, and easier to make mistakes!